

Dr. Elena Rostova

Associate Professor of Computer Science & Data Analytics

Department of Computational Sciences · Apex University

 e.rostova@apex.edu
 elenarostova-academic.org
 Turing Hall, Suite 402
 +1 (555) 019-2834

EXECUTIVE SUMMARY & TEACHING STATEMENT

My pedagogical mission centers on transforming abstract computational theory into empowering, real-world analytical capability. By bridging rigorous computer science fundamentals with inclusive studio-based learning, interactive live-coding, and ethically conscious software engineering, I cultivate students who are not merely proficient coders, but critical thinkers, collaborative problem-solvers, and resilient digital architects. Over nine years of university instruction across undergraduate and graduate levels, I have maintained an average overall teaching evaluation of 4.88 / 5.00, recognized with the Apex University Excellence in Undergraduate Teaching Award (2024).

1. Core Pillars of Teaching Philosophy

My teaching methodology is structured around four interconnected pedagogical pillars designed to engage diverse learner profiles, instill deep conceptual mastery, and foster long-term professional self-efficacy:

Pillar I: Active Learning & Inquiry-Driven Studio Synthesis

Lectures should never be passive transmissions of static information. I structure every course around the **"Studio-Lab" model**, replacing standard multi-hour slide presentations with targeted 15-minute conceptual overviews followed immediately by peer-based collaborative coding sprints and interactive live debugging sessions.

In *CS 201: Data Structures & Algorithms*, for instance, rather than simply writing pseudocode for graph traversal on a whiteboard, students construct visual graph algorithms in real time using custom web tools, deliberately introducing edge cases (e.g., disconnected subgraphs, cyclic loops) to observe algorithmic breakdown. This hands-on inquiry transforms abstract mathematical properties into concrete mental models.

Pillar II: Universal Design for Learning (UDL) & Inclusive Pedagogy

Computational fields often suffer from high barrier perceptions that discourage non-traditional and underrepresented students. I embed Universal Design for Learning principles into all curriculum design by providing **multimodal learning pathways**:

- **Multiple Modes of Representation:** Core concepts are presented simultaneously through formal mathematical proofs, visual architectural diagrams, and runnable open-source codebases.
- **Flexible Assessment Frameworks:** Major project deliverables offer choice in application domain (e.g., climate data analysis, bio-informatics sequencing, or algorithmic music synthesis), allowing students to connect core algorithms to personal intellectual passions.
- **Low-Floor, High-Ceiling Scaffolding:** Introductory assignments begin with intuitive starter code and step-by-step diagnostic checks, while optional extension modules challenge advanced students with research-grade optimization problems.

Pillar III: Industry-Aligned Engineering & Ethical Synthesis

Modern software engineering demands more than syntactical fluency; it requires software hygiene, system architecture awareness, and societal responsibility. In upper-division courses such as *CS 412: Applied Machine Learning*, students operate in simulated agile teams utilizing industry-standard Git version control, continuous integration pipelines, and peer code review workflows.